

The slide features several decorative elements: a light blue hexagon and a dark green hexagon in the top left; a large green hexagon in the top center; and a smaller green hexagon in the bottom center. On the right side, there are overlapping blue geometric shapes, including triangles and polygons, creating a modern, abstract background.

Airbnb Hotel Booking Analysis

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[Internship ID:INTERNSHIP_172663295366ea53f910591]

PROBLEM STATEMENT

Airbnb Hotel Booking Analysis

Airbnb has revolutionized the hospitality industry by connecting travelers with diverse lodging options worldwide. With its expanding listings and user base, analyzing booking data is essential for hosts to optimize earnings and for Airbnb to improve user experience. This project focuses on studying booking patterns, pricing, guest preferences, and host performance to uncover insights that enhance decision-making and guest satisfaction.



Project Description

This project aims to analyze Airbnb booking data to uncover trends and insights that can benefit hosts, guests, and the platform itself. By examining booking behaviors, pricing strategies, guest preferences, and host performance, the analysis seeks to improve decision-making, maximize host earnings, and enhance the guest experience.

Key Objectives:

1. **Booking Patterns** – Identify peak seasons, regional booking variations, and average lead times.
2. **Pricing Strategies** – Analyze effective pricing methods, the impact of seasonality, and how pricing affects guest satisfaction.
3. **Guest Preferences** – Discover the most desired amenities and factors influencing high ratings.
4. **Host Performance** – Examine what top hosts do well, the role of responsiveness, and common guest complaints.

Future Scope:

Future Directions: Develop advanced predictive models, personalize guest recommendations, implement dynamic pricing strategies, and integrate external data sources.

The project can be expanded in the future through:

- **Predictive Analytics** : forecast booking trends and pricing using machine learning.
- **Personalized Recommendations**: for guests based on past behavior.
- **Dynamic Pricing Models**: adjust rates in real time based on market demand and competition.
- **Geospatial Analysis**: examine how location impacts bookings and guest preferences.
- **External Data Integration**: events, weather, and economic trends to improve analysis accuracy.
- **Behavioral Analysis**: study long-term host and guest behavior trends and identify shifts in the hospitality market.

WHO ARE THE END USERS?

- ☐ Airbnb Hosts
- ☐ Airbnb (Platform)
- ☐ Prospective Hosts/Property Investors
- ☐ Guests (Travelers)
- ☐ Data Analysts & Researchers
- ☐ Tourism & Hospitality Stakeholders

Technology Used

Programming Languages

- **Python:** data analysis, visualization

Libraries

- **Pandas & NumPy:** Data manipulation and preprocessing
- **Matplotlib & Seaborn:** Data visualization
- **Plotly Express:** interactive and dynamic charts

Platform: Google Colab

File handling libraries: openpyxl (for Excel) and built-in CSV handling



RESULTS1

Airbnb Hotel Booking Analysis.ipynb

File Edit View Insert Runtime Tools Help

Commands + Code + Text ▶ Run all

[11] ✓ 0s

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import plotly.express as px
```

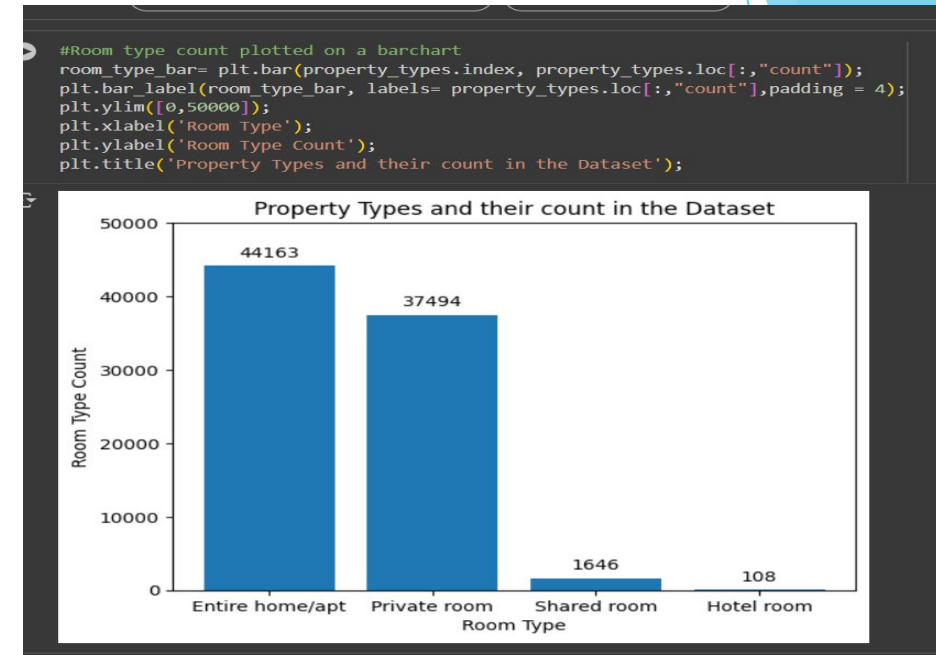
[14] ✓ 0s

```
# Read data from sirbnb dataset
df=pd.read_csv('/content/drive/MyDrive/1730285881-Airbnb_Open_Data.xlsx - in.csv', low_memory=False)
```

[17] ✓ 0s

```
df.head()
```

	id	NAME	host id	host_identity_verified	host name	neighbourhood group	neighbourhood	lat	long	country	...	service fee	minimum nights	number of reviews	last review	review score
0	1001254	Clean & quiet apt home by the park	800144485718	unconfirmed	Madaline	Brooklyn	Kensington	40.64749	-73.97237	United States	...	\$193	10.0	9.0	10/19/2021	0
1	1002102	Skyfit Midtown Castle	52335172823	verified	Jenna	Manhattan	Midtown	40.75362	-73.98377	United States	...	\$28	30.0	45.0	5/21/2022	0
2	1002403	THE VILLAGE OF HARLEM...NEW YORK!	788229239556	NaN	Elise	Manhattan	Harlem	40.80902	-73.94190	United States	...	\$124	3.0	0.0	NaN	N
3	1002755		85098326012	unconfirmed	Garry	Brooklyn	Clinton Hill	40.68514	-73.95976	United States	...	\$74	30.0	270.0	7/5/2019	4



Flows & 20 columns

[72] ✓ 0s

```
#Drop duplicate records
df.drop_duplicates (inplace =True)

#drop house rules and license columns with insufficient data
df.drop(['house_rules', 'license'], axis =1, inplace=True)

#Rename the price and service fee columns to include a dollar sign
df.rename(columns={
    'price': 'price_$',
    'service fee': 'service_fee_$'
}, inplace =True)

# Convert columns to string type before string operations
df['price_$'] = df['price_$'].astype(str)
df['service_fee_$'] = df['service_fee_$'].astype(str)

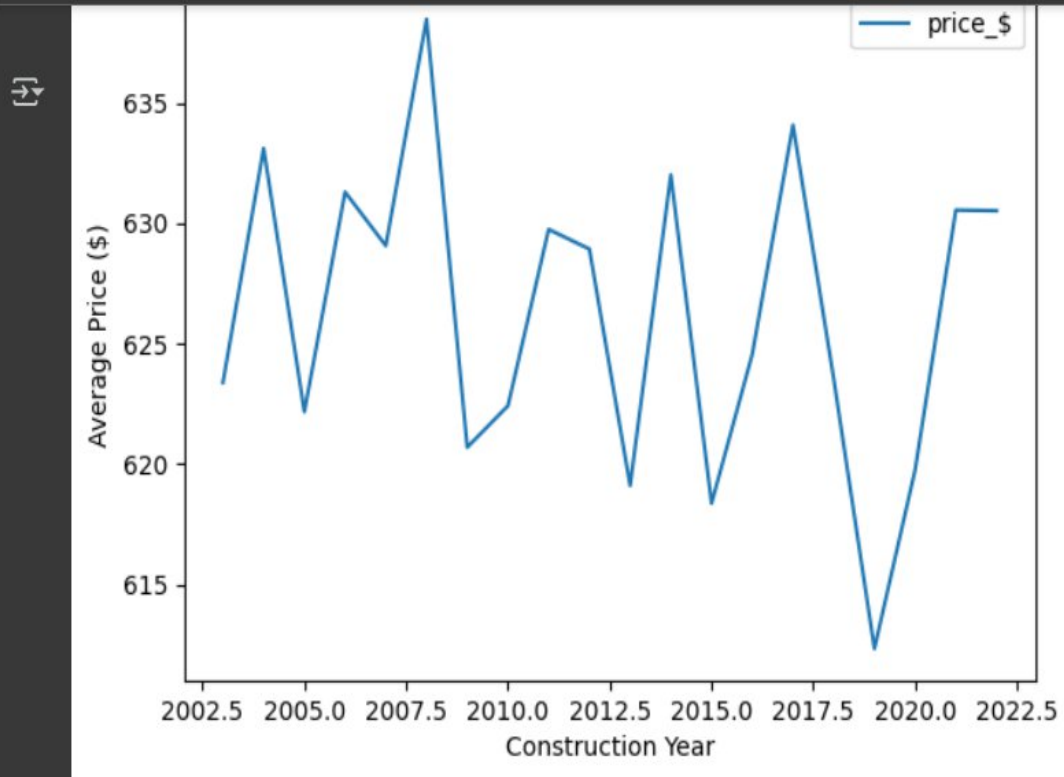
#Remove all dollar signs in the price and service fee columns
df['price_$'] =df['price_$'].str.replace('$','', regex= False)
df['service_fee_$']=df['service_fee_$'].str.replace('$','', regex=False)

#Remove all commas in the price and service fee columns
df['price_$']=df['price_$'].str.replace(',', '', regex =False)
df['service_fee_$'] =df['service_fee_$'].str.replace(',', '', regex=False)

#Drop all records with missing values
df.dropna(inplace = True)

#Change all mismatched data types to the appropriate once.
df['price_$'] = df['price_$'].astype(float)
df['service_fee_$'] =df['service_fee_$'].astype(float)
df['id'] = df['id'].astype(str)
df['host_id']= df['host_id'].astype(str)
```

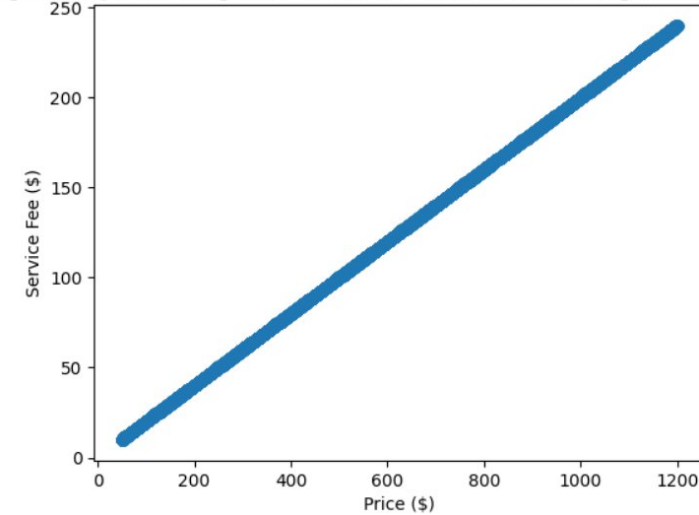
RESULTS2



```
np.float64(0.9999909074778258)
```

```
sns.regplot(df, x='price_$', y='service_fee_$');
plt.xlabel('Price ($)');
plt.ylabel('Service Fee ($)');
plt.title('A regression plot showing the Correlation of the Price of a Listing and its Service Fee');
```

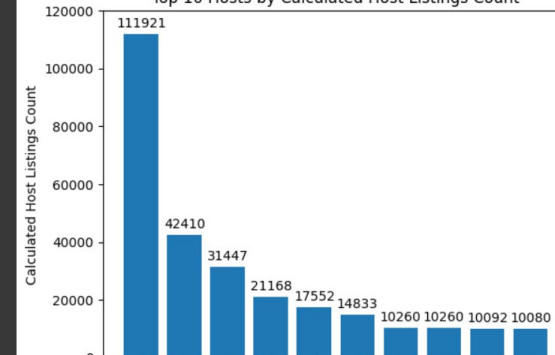
A regression plot showing the Correlation of the Price of a Listing and its Service Fee



```
#Who are the top 10 hosts by calculated host listing count?
hosts= df.groupby('host name')['calculated host listings count'].sum().sort_values(ascending=False).nlargest(10).to_frame()
hosts_bar= plt.bar(hosts.index, hosts.loc[:, 'calculated host listings count']);
plt.bar_label(hosts_bar, label= hosts.loc[:, "calculated host listings count"], label_type='edge', padding = 3);
plt.xlabel('Hosts Name');
plt.ylabel('Calculated Host Listings Count');
plt.xticks(rotation=80);
plt.ylim([0,120000]);
plt.title('Top 10 Hosts by Calculated Host Listings Count');
```

```
#Who are the top 10 hosts by calculated host listing count?
hosts= df.groupby('host name')['calculated host listings count'].sum().sort_values(ascending=False)
hosts_bar= plt.bar(hosts.index, hosts.loc[:, 'calculated host listings count']);
plt.bar_label(hosts_bar, label= hosts.loc[:, "calculated host listings count"], label_type='edge', padding = 3);
plt.xlabel('Hosts Name');
plt.ylabel('Calculated Host Listings Count');
plt.xticks(rotation=80);
plt.ylim([0,120000]);
plt.title('Top 10 Hosts by Calculated Host Listings Count');
```

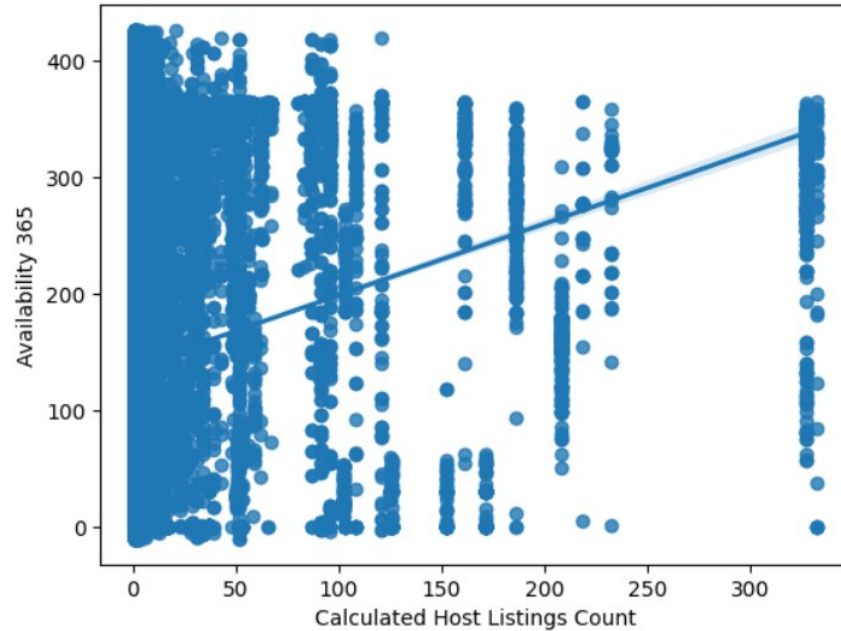
Top 10 Hosts by Calculated Host Listings Count



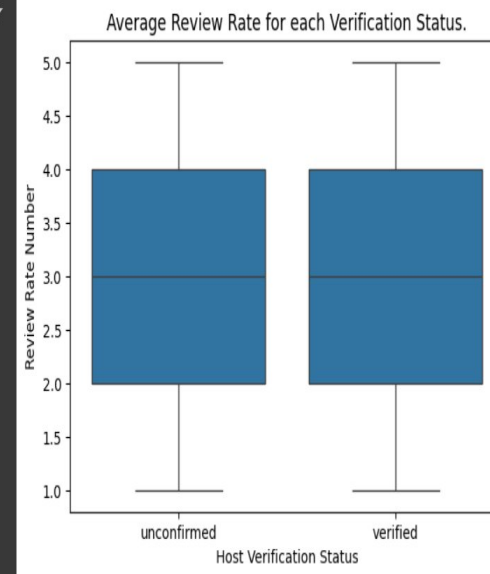
RESULTS3

```
plt.title('A regression plot showing the Correlation of the Calculated Host Listings Count and its Availability 365
```

A regression plot showing the Correlation of the Calculated Host Listings Count and its Availability 365



```
plt.title('Average Review Rate for each Verification Status.');
```



```
#Is there a correlation between the price of listing and its service fee?
```

```
df['price_$'].corr(df['service_fee_$'])
```

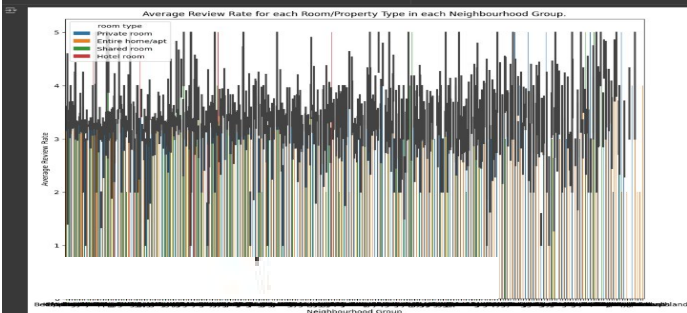
```
np.float64(0.9999909074778258)
```

```
sns.regplot(df, x='price_$', y='service_fee_$');  
plt.xlabel('Price ($)');
```

```
df['calculated host listings count'].corr(df['availability 365'])
```

```
np.float64(0.1359855273675869)
```

```
plt.figure(figsize=(12,10));  
sns.regplot(data=df, x='neighbourhood', y='review rate number', hue='room type');  
plt.xlabel('Neighbourhood Group');  
plt.ylabel('Average Review Rate');  
plt.title('Average Review Rate for each Room/Property Type in each Neighbourhood Group.');
```



					ce_fee_\$	minimum nights	number of reviews	last review	reviews per month	review rate number	calculated host listings count	availability 365
					11.000000	83411.000000	83411.000000	83411	83411.000000	83411.000000	83411.000000	83411.000000
mean	40.727353	-73.948500	2012.488509	626.212730	125.243709	7.405366	32.276642	2019-06-08 22:01:01.027922176	1.377584	3.278776	7.032682	141.742612
min	40.504560	-74.249840	2003.000000	50.000000	10.000000	-365.000000	1.000000	2012-07-11 00:00:00	0.010000	1.000000	1.000000	-10.000000
25%	40.687650	-73.982110	2007.000000	340.000000	68.000000	2.000000	3.000000	2018-10-26 00:00:00	0.220000	2.000000	1.000000	6.000000
50%	40.721320	-73.953720	2012.000000	625.000000	125.000000	3.000000	11.000000	2019-06-13 00:00:00	0.740000	3.000000	1.000000	101.000000
75%	40.762595	-73.930815	2017.000000	914.000000	183.000000	5.000000	38.000000	2019-07-05 00:00:00	2.010000	4.000000	2.000000	266.000000
max	40.916970	-73.705220	2022.000000	1200.000000	240.000000	5645.000000	1024.000000	2022-05-21 00:00:00	90.000000	5.000000	332.000000	426.000000
std	0.056324	0.050347	5.760870	331.787179	66.360979	28.102221	51.957777	NaN	1.751041	1.283599	29.551590	133.368399

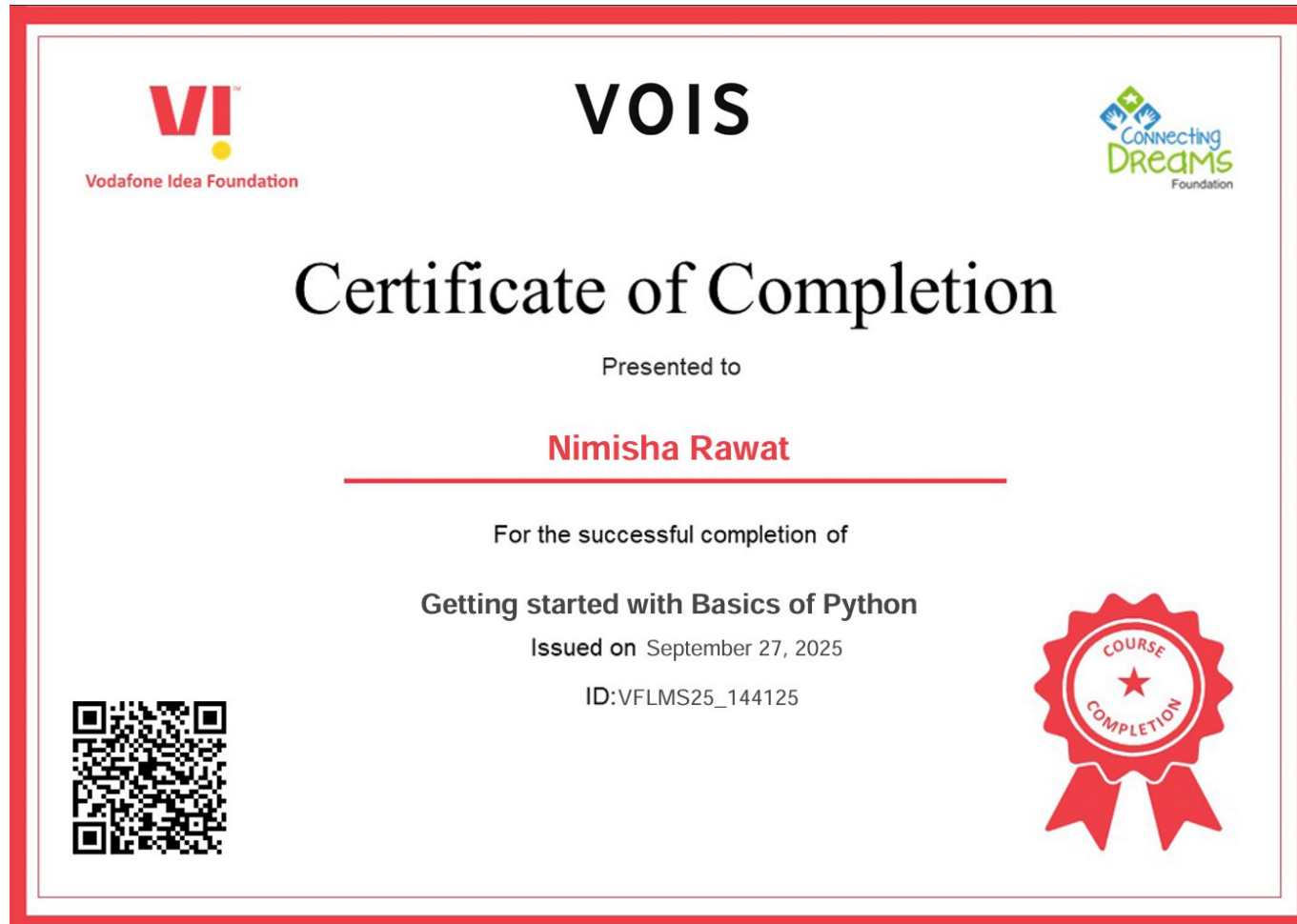
[44]
✓ Os

```
#What are the different property types in the dataset?  
property_types = df['room type'].value_counts().to_frame()
```

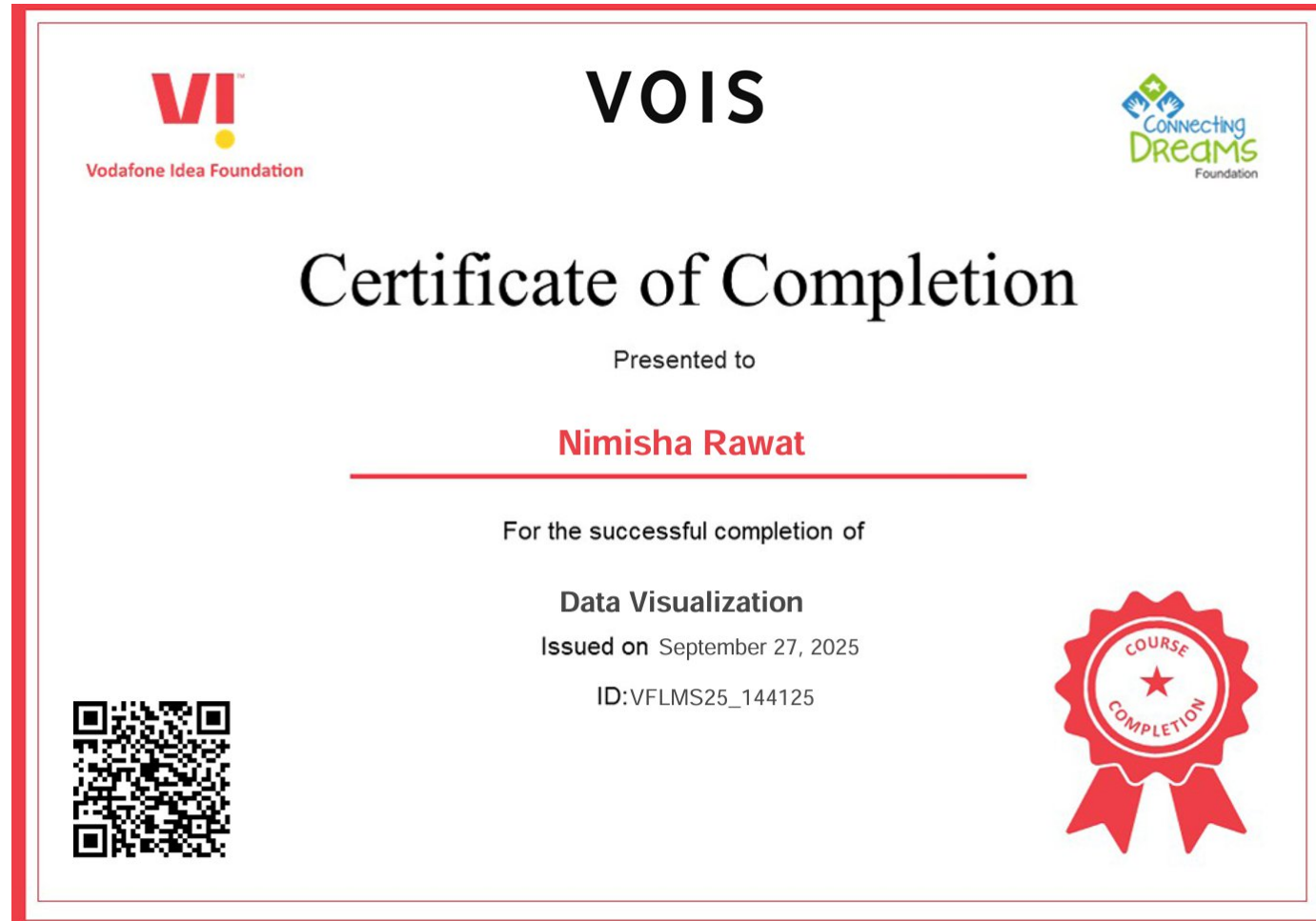

GitHub repository

https://github.com/Nimisha2136/VOIS_AICTE_Oct2025_Nimisha-Rawat.git

Getting started with Basics of Python Certificate



Data Visualization Certificate



Thank you